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SKG 12 Mechanical Engineering

Predictive Maintenance Technique – Part 1 (Failure Mechanism), L2

Failure Mechanisms and Detection Parameters (2 of 4)

Static Equipment

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Failure Mechanisms and Detection Parameters for Static Equipment

Learning Objectives

By the end of this session, you should be able to:

- Understand the various degradation mechanisms affecting equipment components i.e. RCFA approach
- Understand the failure physics relating the degradation parameters for specific failure mechanism i.e. the deterministic approach, to be used in RCFA and detecting incipient failures through Predictive Maintenance Techniques.
- Aware of the re-engineering and RCFA approach that can be implemented to design out or avoid recurrence of failure.
- Aware to the various technology for detection of degradation parameters, and interpretation of results.

Failure Mechanisms related following Static Equipment Covered

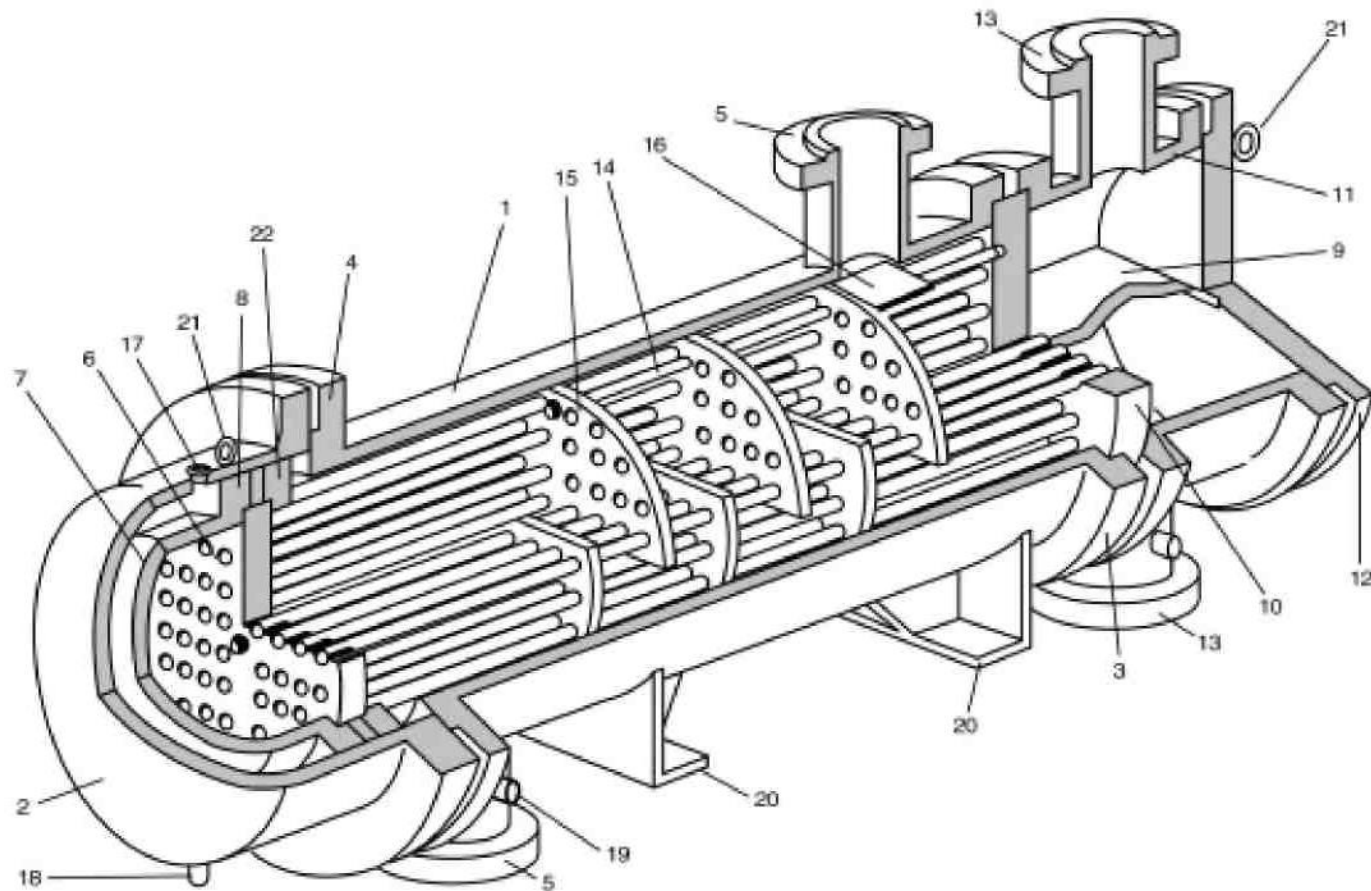
- Heat Exchanger
- Pressure Vessels

Pre-requisite Information

Technical drawings/diagrams containing process data, material and media codes:

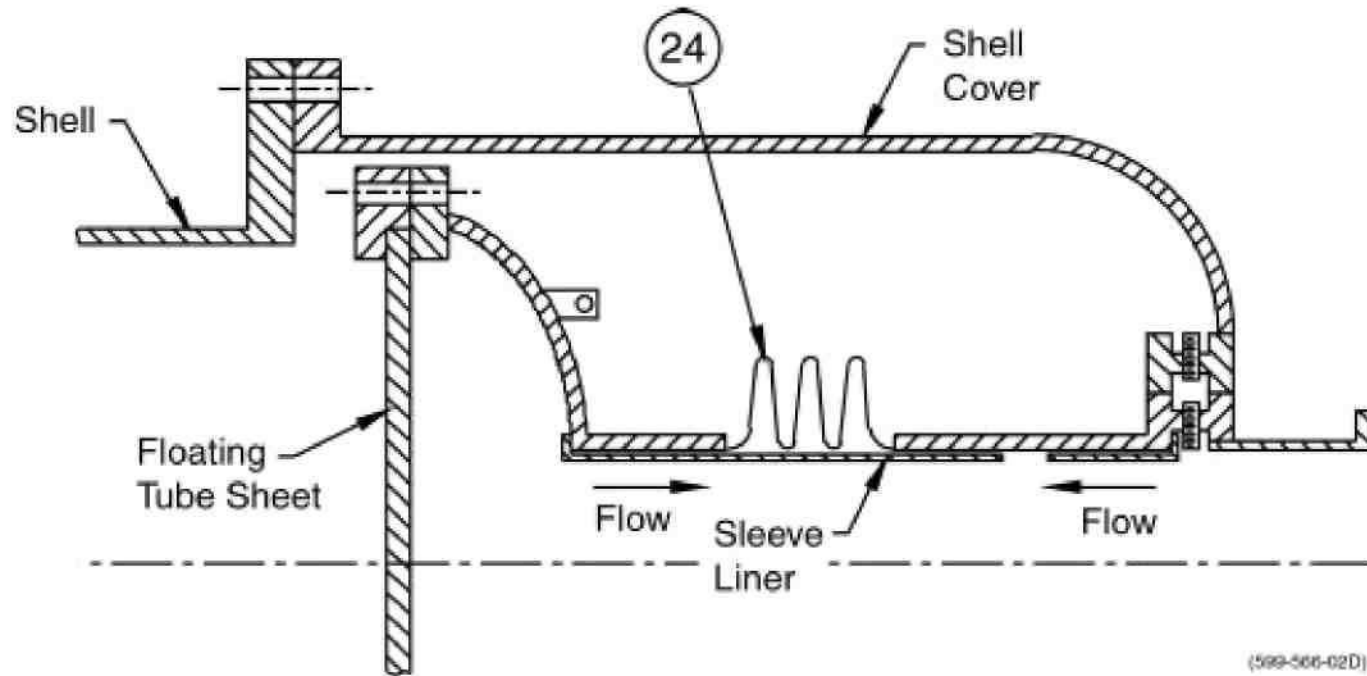
- PFD & P&IDs, Equipment Data Sheet
- Risk Based Inspection (RBI) report
- Shut down logic if any
- Cause and effect diagrams if applicable
- Fire protection data sheets if applicable
- Asset Structure / Acceptance Criteria

Heat Exchangers



- | | | |
|-------------------------------|--|----------------------|
| 1. Shell | 9. Channel Partition Plate | 17. Vent Connection |
| 2. Shell Cover | 10. Stationary Tubesheet | 18. Drain Connection |
| 3. Shell Flange | 11. Channel | 19. Test Connection |
| 4. Shell Cover End Flange | 12. Channel Cover | 20. Support Saddles |
| 5. Shell Nozzle | 13. Channel Nozzle | 21. Lifting Ring |
| 6. Floating Head Tubesheet | 14. Tie Rods and Spacers | 22. Split Ring |
| 7. Floating Head Cover | 15. Transverse Baffles or Support Plates | 23. Tubes |
| 8. Floating Head Cover Flange | 16. Impingement Baffle | 24. Bellows |

Heat Exchangers



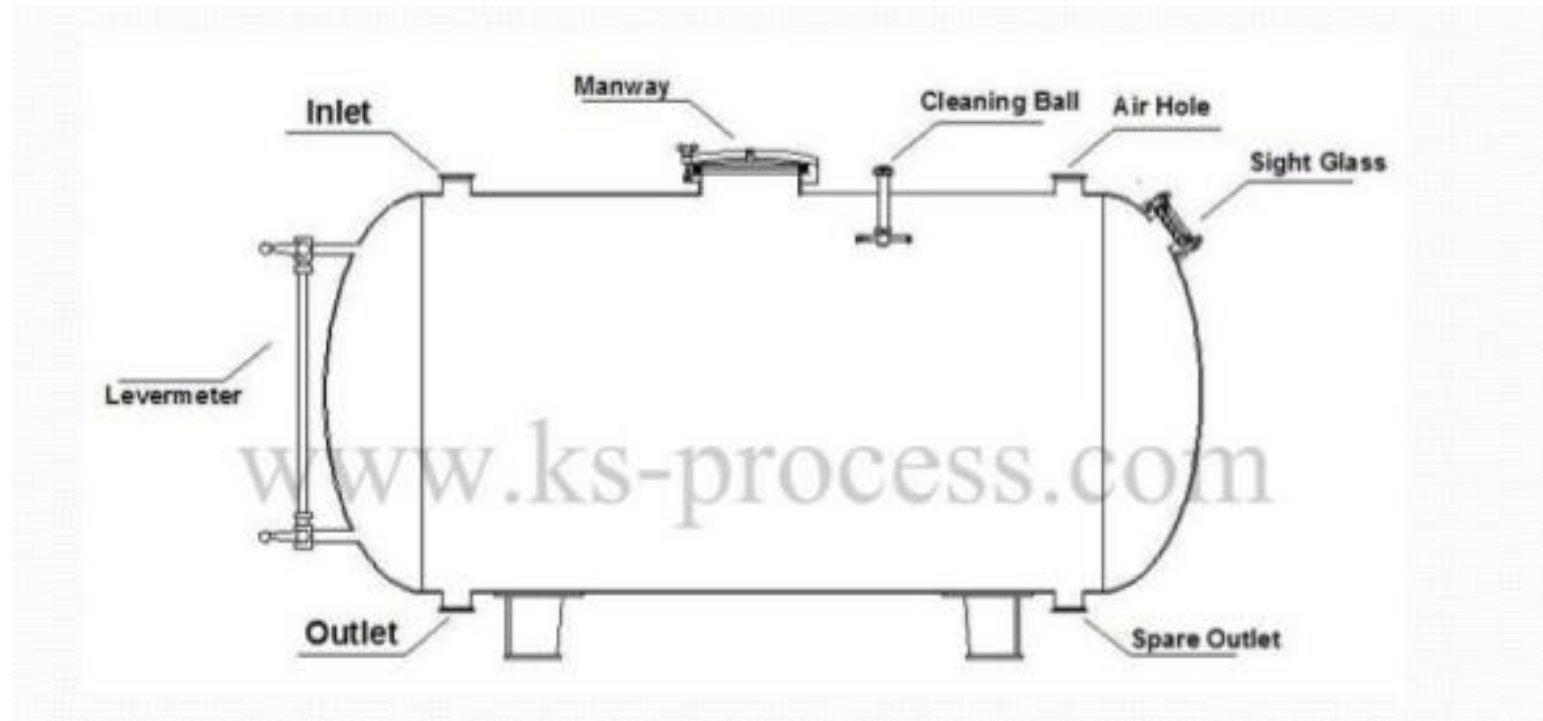
Heat Exchangers

Sno	Failure Mode	Failure Mechanism	Detection Parameter
01	Deterioration of thermal performance and/or increase of pressure drop	Exchanger Fouling (Tube Side) Fouling forms on the tube side of a heat exchanger tube.	Condition Monitor to maintain high tube side velocities.
02	Deterioration of thermal performance and/or increase of pressure drop	Exchanger Fouling (Shell Side) Fouling forms on the shell side of a heat exchanger tube.	Condition monitor to maintain high shell side velocities.
03	Leaking of process medium	Corrosion	Wall thickness measurement

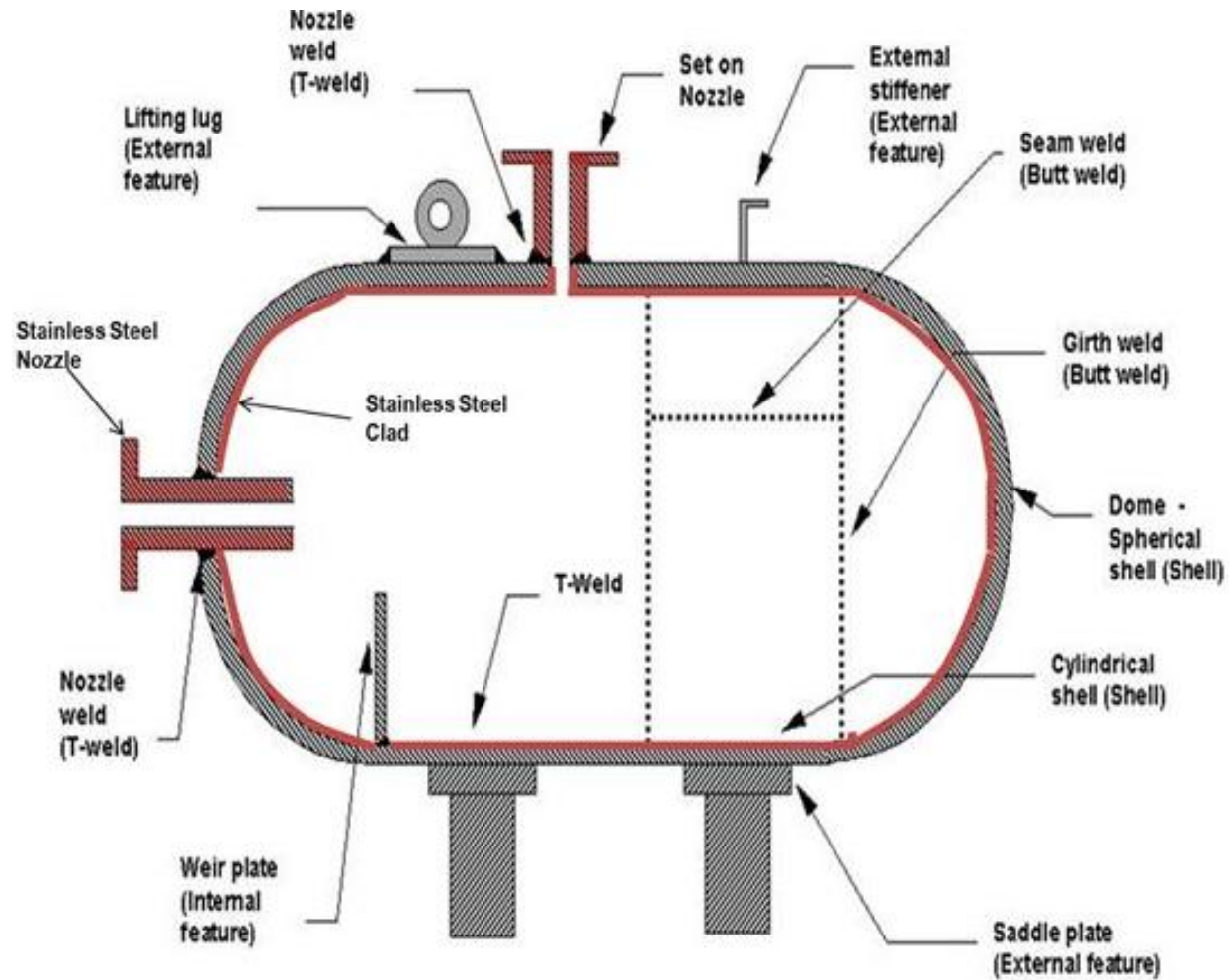
Heat Exchangers Corrosion



Vessels



Vessels



Vessels

Sno	Failure Mode	Failure Mechanism	Detection Parameter
01	Leaking of process medium	Corrosion	Wall thickness measurement
02	Leaking of process medium	Corrosion	Visual Inspection of External
03	Deterioration of thermal performance/ Leaking of process medium	Corrosion under insulation	Perform Assessment of Insulation

Pressure Vessels Corrosion



Pressure Vessels Corrosion under insulation





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Thank you